Fawtir mobile: onboarding to invoice creation, aligned with the web app

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Purpose

The mobile app must collect the same data, apply the same rules and call the same APIs as the Fawtir web app, from sign-up to a filed invoice. This guide lists what to match. It says nothing about screens or layout: those stay your call.

- **Source of truth for rules:** the published PINT-OM specification (docs.peppol.eu/poac/om). Rule IDs are cited so you can check each one.
- **Source of truth for behaviour:** the web app on `main` (repo `fawtir-web`). File paths are given where a list or rule lives, so you can reuse the same data.
- **Money:** always three decimals, and shown with the Omani rial sign, never "OMR" beside a figure.

1. Onboarding

Sign-up is anonymous: every call carries the `onboardingToken` from email verification, and `POST /v1/auth/signup/status` returns the `stage` to resume from. Steps in order:

Step	Fields and rules	API
Sign up	fullName (2 to 200, no digits), username `^[A-Za-z0-9][A-Za-z0-9._-]{2,63}$`, email, phone (+968 then 8 digits starting 2, 7 or 9), password (12 to 128, upper, lower, digit, symbol), consent	`POST /v1/auth/signup`, then `POST /v1/auth/signup/verify` (6-digit code, 10 min) returns `onboardingToken`; `/resend` after 60 s

Step	Fields and rules	API
About business	businessSize: `JUST_ME`, `PEOPLE_2_TO_10`, `PEOPLE_11_TO_50`, `PEOPLE_50_PLUS`. invoicingMethod: `PAPER_OR_MANUAL`, `SPREADSHEETS`, `ACCOUNTING_SOFTWARE`, `FULL_ERP_SYSTEM`. **POS box**: "Will you use Fawtir POS?" yes/no, off by default	`POST /v1/auth/signup/business-profile`
Business type (POS = yes only)	Category; only Food & Beverage is open (`FOOD_AND_BEVERAGE`), others show "not open yet"	`POST /v1/auth/signup/business-category`
Sub type (POS = yes only)	`COFFEE_SHOP`, `RESTAURANT`, `BAKERY`, `FAST_FOOD`, `JUICE_BAR`, `DESSERT_SHOP`, `CATERING`	`POST /v1/auth/signup/business-type`
Business identity	legalCompanyName (max 255), CR number (max 20, upper-cased), **business activity** (below), VATIN `^OM(11|12)\d{8}$` (upper-cased), **Peppol ID** read only (below), invoice accent colour	`POST /v1/auth/signup/business-identity`
Address and contact	addressLine (max 140), postalCode (optional, 3 digits), city, governorate (11 names, sent in English), contact name, role (optional), phone, email	`POST /v1/auth/signup/address-contact`
Review and submit	Read back of all answers	`POST /v1/auth/signup/final-submission` (204), then the pending screen

- **POS = no:** the web app still posts `FOOD_AND_BEVERAGE` and `RESTAURANT` as placeholders, then skips straight to Business identity. The backend has no POS field yet, so the answer is not stored (INCOME.md S.1).
- **Business activity:** a searchable dropdown over the PINT-OM ISIC list, 1,819 six-digit codes with English and Arabic names (`src/lib/codeLists/isic.ts` ; `000000` = Other). Search matches the name or the code. Pre-filled from the sub type (coffee shop `563001` , restaurant `561001` , fast food `561002` , juice bar `563002` , dessert shop `561003` , catering `562901`). Optional at sign-up; sent as `industrialClassificationCode` .
- **Peppol Participant ID:** never typed and never sent. Show `0248:` + the VATIN once the VATIN is valid, else "Filled in from your VAT number". 0248 is Oman's scheme.
- **409 on identity:** "A business with this VATIN or CR number is already registered."

2. Settings › Company profile

The business activity dropdown here is the same ISIC list as at sign-up. It is always editable, and B2B and B2G invoices are refused without it (`INDUSTRIAL_CLASSIFICATION_REQUIRED`).

What	Rule	API
Business activity (ISIC)	Searchable dropdown, 1,819 codes, saved as the 6-digit code. Never asked on the invoice: the backend copies it onto every line (IBR-081-OM)	`GET/PUT /v1/company-profile/industrial-classification` with `{code}` ; needs `admin.settings.manage`
Identity: legal name, Arabic name, trading name, CR, address, city, postal code, country, phone, email	Editable while `verificationStatus` is `PENDING_VERIFICATION` or `REJECTED` ; read only once `VERIFIED`	`GET/PUT /v1/company-profile` ; PUT sends the whole profile plus `expectedRevision`
VAT number	Always read only (changing it means re-registering on Peppol)	same
Brand colour, receipt logo	Always editable; logo is a multipart `file`	`GET/PUT /v1/company-profile/branding` , `POST/DELETE /v1/company-profile/branding/logo`

What	Rule	API
Invoice defaults	language `AR`, `EN` or `AR_EN`; currency OMR only; payment terms; footer note	`GET/PUT /v1/company-profile/invoice-defaults`

The web app saves in this order: profile (only if identity changed and not verified), industrial classification, branding, invoice defaults.

3. Income › Invoice: header fields

The customer type decides almost everything else: B2B and B2G get a full tax invoice over Peppol with line classification, B2C gets a simplified invoice with a QR code and no classification.

Field	Values and rules	Sent as
Customer type	Business (B2B), Consumer (B2C), Government (B2G)	`details.transactionType`: `SIMPLIFIED` for consumer, else `FULL_TAX`
Customer	B2B/B2G: pick from `GET /v1/customers`. B2C: a name plus email or phone; the customer is created automatically (needs `customer.manage`)	`customerId`
Branch	Required; defaults to the head office branch from `GET /v1/branches`	`branchId`
Invoice number	Not entered; numbered on issue	none
Issue date	Required; defaults to today (Muscat); not in the future; in the current year	on issue only

Field	Values and rules	Sent as
Payment terms	`NET0` Due on receipt (B2C default), `NET15`, `NET30` (B2B/B2G default), `NET60`, `EOM` End of month, `ADVANCE`	`details.paymentTerms {code, label}`
Due date	Required; follows the terms until edited; not before the issue date	on issue only
Delivery (supply) date	Optional; not after the issue date	`details.deliveryDate`
Currency	OMR only	`currency: "OMR"`
Default VAT category	Default `S5`; changing it re-sets every line still on the old default	per line (section 5)
Prices entered	**Excl. VAT | Incl. VAT** toggle, default Excl. Incl. is an entry mode only: prices are always saved excluding VAT	`vatMode: "EXCLUSIVE"`
Document discount	Percentage or fixed OMR amount, default 0; applied after line discounts, shared across lines by value, before VAT	`headerDiscount {type: PERCENTAGE | FIXED_AMOUNT, value}`
PO reference	Optional, max 200	`details.purchaseOrderReference`
Subject	Optional, max 500	`details.subject`
Note to customer / internal note	Optional, max 2,000 each	`details.customerVisibleNote`, `details.internalNote`
Business activity	B2B/B2G only: shown read only ("code · name") from the company profile; warn if missing	not sent

Field	Values and rules	Sent as
Payment method	UNCL4461: `10` Cash, `20` Cheque, `30` Bank transfer, `48` Card, `68` Online	not in the draft yet

4. Income › Invoice: line fields

Every line is free text with a Goods/Service switch. The switch drives the unit list and the classification dropdown.

Field	Values and rules	Sent as
Description	Free text, required, max 500. A catalogue pick may fill description, unit, price and VAT	`description` (+ `productId`, `productCode` if picked)
Goods / Service	Default Goods. Switching resets the unit if it is not on the new list and clears the other kind's classification	`classification.goodsOrService`: `G` or `S` (IBR-078-OM)
Unit (goods)	EA Each (default), H87 Piece, KGM, GRM, TNE, LTR, MLT, MTR, CMT, MTK, MTQ, BX, CT, PK, BG, BO, CA, DZN, SET, PR, RO, **ZZ Other** (last)	`unitCode` (UN/ECE Rec 20)
Unit (service)	HUR Hour, DAY, WEE, MON, ANN, E48 Job, C62 Unit, LS Lump sum, KMT, **ZZ Other** (last)	`unitCode`
Quantity	Greater than 0	`quantity`
Price excl.	Three decimals (normalised on blur), not negative; qty × price at most 999,999,999,999.999	`unitPrice` (always excl. VAT)
Line discount	Percentage (default) or fixed OMR; a percentage also shows its OMR amount	`discount {type, value}`

Field	Values and rules	Sent as
VAT category	Per line; starts from the header default; changing it clears the reason	`taxCategory {code, en16931Category, rate}`
Reason	Only for Z, E and O (section 5)	`exemptionReasonCode` or `exemptionReason`
Classification	**B2B and B2G only**, hidden for B2C	see below

HS code (goods lines):

- Searched on the server: `GET /v1/code-lists/hs?q=&limit=20` over 14,313 published 12-digit codes, returning `{items:[{code, name, nameAr}], total}`. The web debounces 250 ms.
- Always offer **Other =** `000000000000` at the end of the list.
- Required on every goods line of a B2B/B2G invoice (IBR-079-OM); must be exactly 12 digits (IBR-080-OM). An unknown code is refused with `HS_CODE_UNKNOWN`.
- Sent as `classification: {goodsOrService: "G", hsCode}`.

Service type (service lines):

- 21 published CL-12-OM codes, filtered on the device (`src/lib/codeLists/serviceTypes.ts`). **Other =** `00000000`.
- Only meaningful for an export of services (Z with `VATZR-OM-09`, IBR-155-OM). Not required otherwise: send `{goodsOrService: "S"}` and the backend fills `00000000`.

Business activity (ISIC) is never asked on the line; the backend copies it from the company profile.

5. VAT categories and reasons

Oman allows exactly four VAT categories (CL-04-OM). Load them from `GET /v1/reference-data` (`taxCategories`) and keep only S, Z, E and O. The server sets the rate from the category and ignores the rate you send.

Code	Label	Rate	Reason needed	Sent as
`S5`	Standard rate	5%	None: no code or text (ALIGNED-IBRP-S-10-OM)	nothing
`Z0`	Zero-rated	0%	A code from **VATZR-OM-01 to 16** (CL-10-OM, IBR-069-OM)	`exemptionReasonCode`
`E0`	Exempt	0%	A code from **VATEX-OM-01 to 12** (CL-05-OM, IBR-069-OM)	`exemptionReasonCode`
`O0`	Out of scope	0%	Free text, max 500; never a code (IBR-070-OM)	`exemptionReason`

- Lines on one invoice may mix categories, including O with the others.
- An export is **Z** with VATZR-OM-09 to 12, not a separate category. Reverse charge is a transaction type, not a category.
- The full code labels, in English and Arabic, are in `src/lib/codeLists/taxReasons.ts`. For example: VATZR-OM-01 qualifying food, 02 medicines and medical equipment, 09 export of service, 10 direct export of goods; VATEX-OM-01 financial services, 02 education, 03 healthcare, 04 residential rent, 05 local passenger transport.
- Missing reason: `EXEMPTION_REASON_REQUIRED` from the server.

Zero-rated food check (coming, backend): for B2B/B2G goods lines, the draft preview will return warnings `ZR_FOOD_NOT_LISTED` (Z with VATZR-OM-01 but the HS code is not on the food list) and `ZR_FOOD_CHARGED_STANDARD` (S on a listed food code). They are warnings, not blocks: issuing then needs `confirmedWarnings`. Service lines and B2C are not checked.

Rules: PINT-OM Oman rules · code lists

6. Totals, prices and rounding

Show a live estimate while typing, but treat the server preview (`POST /v1/invoice-drafts/{id}/preview`) as the figure of record. The web app's math is in `src/lib/invoiceMath.ts` if you want to port it exactly.

1. Gross = quantity × price excl. VAT.
2. Line discount: a percentage of gross, or a fixed amount; capped between 0 and gross.

3. The document discount is applied next, before VAT. A percentage applies to each line. A fixed amount is shared across lines in proportion to each line's value after its own discount.
4. Taxable = after line discount − its share of the document discount.
5. VAT is worked out per rate on the grouped taxable amount, then shared back to the lines, so lines always add up to the total.
6. Total = subtotal − line discounts − document discount + VAT.

- **Incl. VAT entry:** price excl. = price incl. × 100 ÷ (100 + rate). Flipping the toggle converts every line; 0% lines stay as they are. Always send the excl. price.
- **Rounding:** half up, three decimals on everything shown and sent. Keep more precision inside the calculation.
- **Summary rows:** Subtotal, Line discounts, Document discount, Net amount, VAT, Total.
- **Server refusals:** `LINE_DISCOUNT_EXCEEDS_BASE`, `HEADER_DISCOUNT_EXCEEDS_BASE`, `LINE_AMOUNT_TOO_LARGE`.

7. Bulk invoice upload

The file is taken whole or not at all, and it only ever creates drafts, never issued invoices. The backend endpoints are specified but not live yet: a 404 means "Invoice import is not available yet".

1. **Choose the type:** `BUSINESS` (customers matched by VAT number or name), `CONSUMER` (walk-in, created from a name; needs `customer.manage`) or `GOVERNMENT`. Needs `invoice.draft`.
2. **Download the template:** `GET /v1/invoices/import/template?kind=…&language=en|ar` returns an `.xlsx` generated by the server. It includes an example row and a sheet of valid values.
3. **Upload:** `.xlsx` only, max 10 MB and 2,000 rows. `POST /v1/invoices/import/check` (multipart `file`, `kind`, `language`) returns a batch `{id, state, totalRows, processedRows, errors, summary, inferred, unresolved, duplicateOf}`.
4. **Wait if large:** over 50 rows runs in the background (`state` CHECKING, then CREATING). Poll `GET /v1/invoices/import/{id}` about every 1.5 s and show `processedRows / totalRows`.
5. **Fix errors:** any error blocks the whole file. Each error has a row, a column and a message (for example "KG is not a valid unit. Did you mean KGM"). If `duplicateOf` is set, warn that the same file was already uploaded, with when and by whom.

6. **Preview:** show `summary` (invoices, lines, total, per customer) and `inferred` (values Fawtir filled in). For each `unresolved` description (B2B/B2G), ask once for an HS code or service type with the same classification lists as section 4.
7. **Create:** `POST /v1/invoices/import/{id}/commit` with `{classifications: [{description, itemType, code}]}`. The result lists the created drafts (`reference, invoiceId, draftNumber, customerName, total`); the report is at `GET /v1/invoices/import/{id}/report`. Cancel with `DELETE /v1/invoices/import/{id}`.

Template rules: one row per invoice line; rows sharing a reference in the first column make one invoice. Invoice numbers, dates and IDs are generated by Fawtir. A blank unit means EA for goods or C62 for services, and a blank branch means head office. OMR only. Every non-standard VAT line needs its reason (section 5).

8. APIs to wire

Onboarding and company profile calls are listed in sections 1 and 2; bulk upload in section 7. The invoice flow:

Step	Call	Notes
Reference data	`GET /v1/reference-data`	VAT categories, units
Branches, customers	`GET /v1/branches` , `GET /v1/customers`	
HS search	`GET /v1/code-lists/hs?q=&limit=20`	server search
Create draft	`POST /v1/invoice-drafts`	header `Idempotency-Key: draft-<uuid>` ; body `{type: "INVOICE", draft}`
Update draft	`PUT /v1/invoice-drafts/{id}`	`{expectedVersion, draft}` ; `STALE_INVOICE_VERSION` on conflict
Load, discard	`GET` / `DELETE /v1/invoice-drafts/{id}?expected_version=`	

Step	Call	Notes
Preview totals	`POST /v1/invoice-drafts/{id}/preview`	amounts at 3 decimals
Issue	`POST /v1/invoice-drafts/{id}/issue`	`Idempotency-Key`; `{expectedVersion, issueDate, dueDate}`; issuing does not send

Draft body (unknown fields are refused):

```
{
  "customerId": "…", "branchId": "…", "currency": "OMR", "vatMode":
"EXCLUSIVE",
  "headerDiscount": { "type": "PERCENTAGE", "value": 0 },
  "details": {
    "transactionType": "FULL_TAX", "paymentTerms": { "code": "NET30",
"label": "Net 30 days" },
    "deliveryDate": null, "subject": null, "purchaseOrderReference": null,
"customerVisibleNote": null
  },
  "lines": [{
    "lineNumber": 1, "description": "Arabic coffee beans 1 kg", "unitCode":
"KGM",
    "quantity": 2, "unitPrice": 6.5, "discount": { "type": "PERCENTAGE",
"value": 0 },
    "taxCategory": { "code": "S5", "en16931Category": "S", "rate": 5,
"exemptionReason": null, "exemptionReasonCode": null },
    "classification": { "goodsOrService": "G", "hsCode": "090121000000" }
  }]
}
```

The HS code above is only an example; look it up with the HS search call.

Issue refusals to handle: `INDUSTRIAL_CLASSIFICATION_REQUIRED`, `SELLER_PARTICIPANT_NOT_VERIFIED`, `COMPANY_VERIFICATION_PENDING`, `QUOTA_EXCEEDED`, `PERIOD_CLOSED`, `STALE_INVOICE_VERSION`. The full list and messages are in `src/lib/invoiceIssue.ts`.

Also used around the invoice flow:

Purpose	Call	Notes
Sign-up resume	`POST /v1/auth/signup/status`	`{onboardingToken}`; returns `stage` and which steps are complete
Resend sign-up code	`POST /v1/auth/signup/resend`	`{tenantCode, email}`; always 202
Business activity search (optional)	`GET /v1/code-lists/isic?q=`	server search over the same ISIC list; the web bundles the list instead
Customer list, one customer	`GET /v1/customers`, `GET /v1/customers/{id}`	
Create a customer (B2C walk-in too)	`POST /v1/customers`	needs `customer.manage`
Catalogue search, add a product	`GET /v1/products?q=&size=`, `POST /v1/products`	a pick fills description, unit, price and VAT
Internal note	`GET /v1/authored-invoices/{id}/internal-note`	needs `invoice.internal-note.read`
Attachments	`GET` / `POST /v1/authored-invoices/{id}/attachments`, `GET` / `DELETE …/attachments/{attachmentId}`	upload after the draft is saved; `.pdf .jpg .png .xlsx`, max 10 MB

Sending over Peppol has no endpoint yet: issuing does not send, and delivery status is limited to `NOT_SUBMITTED` or `SENT_MANUALLY` for now.

Signing in once the company is activated (between onboarding and the first invoice):

Purpose	Call	Notes
Sign in	`POST /v1/auth/login`	`{username, password}`. The web sends `portalCookie: true` and gets a cookie session; mobile needs the token mode, so confirm the body and token handling with the backend. Returns `mfaRequired`, `mfaChallengeToken`, `mfaRequiredFactor`
Ask for the sign-in code	`POST /v1/auth/mfa/otp/request`	email code by default (`EMAIL_OTP`); allow a resend after 60 s
Verify the code	`POST /v1/auth/mfa/verify`	with the challenge token
Authenticator setup (only if the server demands it)	`POST /v1/auth/mfa/totp/enroll`, `POST /v1/auth/mfa/totp/confirm`	when `mfaState` is `ENROLLMENT_REQUIRED`
Who am I, what may I do	`GET /v1/me/entitlements`	`permissions` (for example `invoice.draft`, `customer.manage`, `admin.settings.manage`), roles, tenant name; use it to decide what to offer
Forgot / reset password	`POST /v1/auth/forgot-password`, `POST /v1/auth/reset-password`	
Sign out	`POST /v1/auth/logout`	

With this, every call the web app makes from sign-up to an issued invoice is listed in this doc. The plans and payment screens make no API calls yet.

9. Checklist

- [ ] POS yes/no asked at sign-up; POS = no skips business type and sub type
- [ ] Business activity (ISIC, 1,819 codes, searchable, Other = 000000) at sign-up, pre-filled from the sub type
- [ ] Peppol ID shown read only as `0248:` + VATIN, never typed or sent
- [ ] VATIN checked against `OM11` or `OM12` + 8 digits
- [ ] Company profile: ISIC always editable; identity locked once verified; VAT number always locked
- [ ] Customer type B2B / B2C / B2G sets the transaction type and hides classification for B2C
- [ ] Goods/Service switch per line, with its own unit list ending in ZZ Other
- [ ] HS code search on the server, Other = 000000000000, required on B2B/B2G goods lines
- [ ] Service type list (21 codes), Other = 00000000, not required
- [ ] Price excl. at three decimals; Excl./Incl. toggle converts prices; always send excl.
- [ ] Line discount and document discount, each as percentage or fixed OMR
- [ ] VAT category per line from S5, Z0, E0, O0; the header sets the default
- [ ] Z needs a VATZR code, E a VATEX code, O free text, S nothing
- [ ] Totals from the server preview; three decimals; rial sign, never "OMR"
- [ ] Bulk upload: type, template, `.xlsx` check, errors, preview with unresolved classifications, commit
- [ ] English and Arabic for every label

Open questions for the backend: where the POS answer is stored (S.1); whether O0 must always carry a reason (V.5); when the import and zero-rating food check endpoints go live.